
Paired-Acquisition Neural Factorization for Computational Pathology

A corrected representation-level study of scanner-structured histopathology embeddings

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github.com/matthewvaishnav/computational-pathology-research

Abstract

Frozen histopathology embeddings can encode scanner identity together with tissue structure. Removing every scanner-predictive direction is unsafe when acquisition and tissue are correlated, yet leaving scanner information uncontrolled can compromise transport across devices. We study Paired-Acquisition Neural Factorization (PA-NF), a two-branch model trained from matched acquisitions of the same tissue region. A tissue-oriented branch is optimized for same-region agreement and reduced scanner recoverability; an acquisition branch is optimized to retain scanner information; a joint decoder and variance controls discourage collapse.

The primary corrected analysis uses SCORPION: 48 human H&E slides, 480 aligned regions, five scanners, and 2,400 patches. In a separately versioned 175-fit, capacity-matched campaign, PA-NF reduced held-out linear scanner balanced accuracy by 0.3108 relative to an equal-parameter two-branch control without scanner objectives (95% fold/slide bootstrap interval $[-0.3346, -0.2858]$). Mean top-1 same-region retrieval changed by only $+0.00004$ ($[-0.00009, 0.00026]$), within a preregistered 0.02 noninferiority margin. The acquisition branch remained strongly scanner informative.

On an independent five-scanner canine cutaneous squamous-cell-carcinoma dataset, corrected five-category evaluation yielded a materially more conservative result. Neural B32/B64 representations did not establish a feature-space increment over simple centroid/QR or paired-linear scanner-removal baselines. Increasing the tissue bottleneck from 32 to 64 dimensions increased scanner recoverability by 0.0489 and retrieval by 0.0515, but did not improve corrected category balanced accuracy (-0.0076 , interval $[-0.0194, 0.0071]$). The evidence supports partial structured separation under the tested protocols, not pure biological factors, complete scanner invariance, diagnostic improvement, or clinical utility.

1 Introduction

Digital pathology pipelines inherit variation from tissue preparation, staining, scanner optics, color response, compression, registration, and local workflow. Modern self-supervised and pathology-specific encoders can preserve useful morphology while also making the acquisition device highly recoverable. This creates a problem that ordinary domain invariance does not solve: a direction correlated with scanner may also carry tissue information, so suppressing it unconditionally can remove information that the model should retain.

Matched acquisitions change the identification problem. When the same tissue region is scanned on several devices, within-region differences are candidates for acquisition variation while shared structure is a candidate for tissue-associated content. PA-NF uses that repeated-measures structure to learn two explicit representations rather than treating nuisance removal as unaccounted deletion.

This paper makes four bounded contributions. First, it specifies a two-branch paired-acquisition objective with explicit retention, routing, reconstruction, and collapse

controls. Second, it reports a capacity-matched SCORPION campaign with the original slide as the statistical unit. Third, it evaluates the resulting representation family on an independent five-scanner canine dataset under a corrected, fixed five-category estimand and identical candidate pools. Fourth, it publishes the negative result that the neural representation did not establish an increment over strong simple corrections. The target is an auditable representation study, not a clinical-performance claim.

2 Related Work and Problem Setting

Scanner variability is a known source of domain shift in computational pathology. The SCORPION benchmark provides spatially aligned views of 480 tissue regions scanned on five devices and introduces SimCons for scanner consistency evaluation [1]. The multi-scanner canine cutaneous squamous-cell-carcinoma dataset similarly provides local correspondences across five scanners for 44 biological samples [2]. These datasets make scanner effects observable while controlling more tissue variation than unpaired institutional cohorts. Domain-adversarial training seeks features that confuse a

domain classifier [3]. Such objectives are useful, but low domain predictability alone does not establish retained task information. PA-NF therefore combines scanner suppression with same-region retention, an explicit acquisition branch, reconstruction, and collapse diagnostics. The experiments operate on frozen DINOv2, Phikon, or ResNet50 representations rather than training the image backbone end to end; DINOv2 is a general-purpose self-supervised visual encoder [4], while Phikon is pathology-specific [5].

The scientific target is conditional allocation. Let $x_{r,s}$ denote the embedding of tissue region r acquired by scanner s . Under the illustrative additive model

$$x_{r,s} = Bz_r^* + Az_s^* + \epsilon_{r,s}, \quad (1)$$

paired differences cancel the shared tissue term:

$$x_{r,s'} - x_{r,s} = A(z_{s'}^* - z_s^*) + (\epsilon_{r,s'} - \epsilon_{r,s}). \quad (2)$$

The implemented neural model does not require exact additivity, but the cancellation argument explains why matched acquisition views contain information absent from scanner labels alone.

3 Method

3.1 Architecture

For a standardized frozen embedding $x \in \mathbb{R}^D$, two multilayer perceptrons produce a tissue-oriented factor and an acquisition factor:

$$z_{\text{tissue}} = f_t(x) \in \mathbb{R}^{d_t}, \quad d_t \in \{32, 64, 256\}, \quad (3)$$

$$z_{\text{acq}} = f_a(x) \in \mathbb{R}^{d_a}, \quad d_a = 64, \quad (4)$$

$$\hat{x} = g([z_{\text{tissue}}; z_{\text{acq}}]) \in \mathbb{R}^D. \quad (5)$$

The SCORPION reference configuration uses $d_t = 256$; the canine bottleneck allocation study compares $d_t = 32$ and 64. Both encoders use a 512-unit GELU layer with layer normalization followed by a linear projection. The decoder maps the concatenated factors through width 512 back to the input feature dimension.

A scanner head q_t is attached to z_{tissue} through gradient reversal, while q_a predicts scanner directly from z_{acq} . The tissue branch is therefore trained against scanner recovery, whereas the acquisition branch is trained to retain it. Figure 1 shows the routing logic.

3.2 Training Objective

Let a minibatch be $\{(x_i, r_i, s_i)\}_{i=1}^n$, where r_i is region identity and s_i is scanner identity. With normalized tissue factors $u_i = z_{\text{tissue}}^{(i)} / \|z_{\text{tissue}}^{(i)}\|_2$, the supervised paired contrastive term is

$$P(i) = \{p \neq i : r_p = r_i\}, \quad (6)$$

$$\mathcal{L}_{\text{pair}} = -\frac{1}{n} \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(u_i^\top u_p / \tau)}{\sum_{j \neq i} \exp(u_i^\top u_j / \tau)}, \quad (7)$$

with $\tau = 0.1$. Reconstruction uses mean squared error,

$$\mathcal{L}_{\text{recon}} = \frac{1}{nD} \|\hat{X} - X\|_F^2. \quad (8)$$

Variance penalties on both branches prevent constant solutions; a within-tissue covariance penalty discourages redundant coordinates. Scanner cross-entropy is reversed only at the tissue encoder and optimized normally for the acquisition encoder.

A direct dependence term penalizes standardized tissue-factor covariance with one-hot scanner indicators:

$$\mathcal{L}_{\text{dep}} = \frac{1}{d_t C} \left\| \frac{1}{n-1} \tilde{Z}_t^\top \tilde{Y}_s \right\|_F^2. \quad (9)$$

A cross-factor covariance penalty limits linear dependence between the centered tissue and acquisition factors:

$$\mathcal{L}_{\text{xcov}} = \frac{1}{d_t d_a} \left\| \frac{1}{n-1} (Z_t^c)^\top Z_a^c \right\|_F^2. \quad (10)$$

The frozen SCORPION objective is

$$\mathcal{L} = \mathcal{L}_{\text{pair}} + \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{var},t} + 0.25\mathcal{L}_{\text{var},a} + 0.01\mathcal{L}_{\text{cov},t} \quad (11)$$

$$+ 0.5\mathcal{L}_{\text{scan},t} + 0.5\mathcal{L}_{\text{scan},a} + 20\mathcal{L}_{\text{dep}} + 0.05\mathcal{L}_{\text{xcov}}. \quad (12)$$

The objective is evaluated as an implemented design, not as proof that every term is universally necessary.

4 Data and Evaluation

4.1 SCORPION

SCORPION contains 48 H&E slides, each scanned on Leica Aperio AT2, Leica Aperio GT450, Roche Ventana DP200, 3DHitech P1000, and Philips UFS B300 devices. Ten aligned regions per slide yield 480 matched regions and 2,400 patches [1]. Five rotating folds keep every region and scanner view from an original slide in the same fold.

The capacity-matched campaign contains seven variants, five folds, and five seeds: 175 completed fits with no failed or invalid cells. The primary control is an equal-parameter two-branch model without scanner objectives. Secondary ablations remove the acquisition classifier, adversarial scanner head, direct scanner-dependence term, or cross-factor covariance term. Seeds are averaged within fold, variant, and slide before contrasts. Confidence intervals use 100,000 bootstrap draws, resampling folds first and then slides within fold. No slide-independent sign-flip p -values are reported.

4.2 Independent Multi-Scanner Canine SCC Data

The external representation audit uses 44 canine cutaneous squamous-cell-carcinoma samples digitized by five scanners with local correspondences across devices [2]. Biological samples, not patches or views, define the folds. The corrected category estimand retains Dermis, Epidermis, Inflammation/Necrosis, SCC, and Subcutis; Bone and Cartilage are excluded. Each retained category has at least two fit and two held-out biological samples per fold. Standardization and probe fitting use fit rows

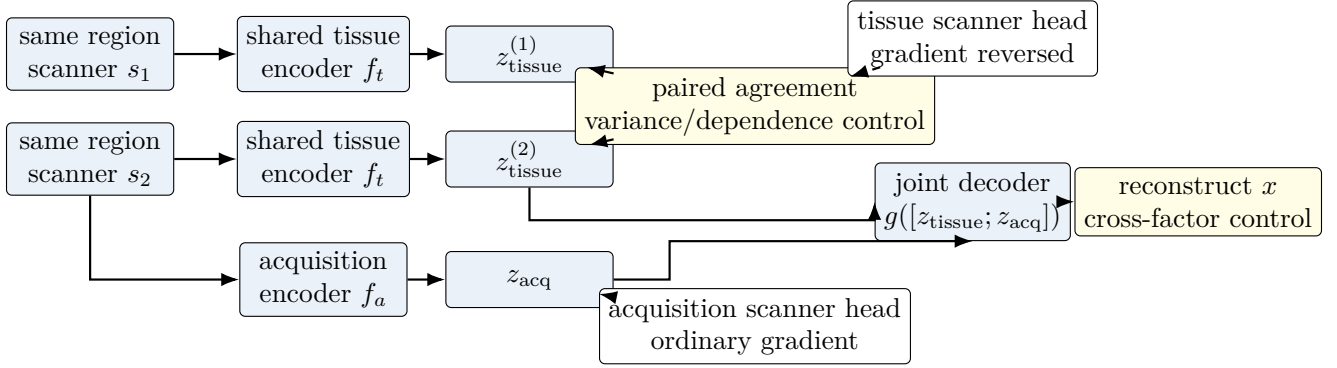


Figure 1: PA-NF routes matched tissue views into a tissue-oriented branch and routes scanner information into an explicit acquisition branch. Reduced scanner probe performance is interpreted only together with retention metrics, acquisition-branch scanner recovery, and collapse controls.

only. Nearest-neighbor candidate pools exclude the same region and the same biological sample where specified. The v2 adjudication loads exact recovered neural representations and trains no new neural models. It compares original frozen features, centroid/QR scanner-subspace projection, a paired linear scanner transform, PCA component removal, a scanner-balanced random control, and neural tissue bottlenecks B32 and B64 under identical candidate pools and probe families.

4.3 Metrics and Decision Rules

Scanner balanced accuracy measures linearly recoverable device identity; lower values are favorable only if retention does not collapse. Tissue retention is summarized by same-region cross-scanner top-1 retrieval and cosine agreement. On canine SCC, descriptive tissue-category balanced accuracy is reported as a representation audit, not as diagnosis or clinical validation.

SCORPION retrieval noninferiority uses a fixed 0.02 margin. Canine adjudication uses three axes: lower scanner balanced accuracy, higher corrected category balanced accuracy, and higher worst-scanner-pair retrieval. A neural feature-space increment requires category balanced accuracy at least 0.02 above every simple baseline in at least four folds without a scanner or worst-pair retrieval regression larger than 0.02. No weighted composite score is used.

5 Results

5.1 Capacity-Matched SCORPION Evidence

Table 1 reports the primary registered comparison. Relative to the equal-parameter two-branch control without scanner objectives, PA-NF reduced tissue-branch scanner balanced accuracy by 0.3108. Mean and worst cross-scanner cosine increased, while top-1 retrieval remained effectively unchanged and inside the preregistered noninferiority margin. The acquisition branch gained 0.118 scanner balanced accuracy relative to the same control, and its absolute scanner accuracy was 0.6565 above five-class chance.

The ablations localize, but do not uniquely identify, the mechanism. Removing the direct scanner-dependence

term increased tissue scanner recoverability by 0.3164 relative to the full objective; removing the acquisition classifier reduced acquisition-branch scanner recovery by 0.0670. Cross-factor covariance and adversarial-head ablations produced smaller or mixed changes. These results support the registered objective as a whole while arguing against claims that each component is independently necessary in every setting.

Pair-integrity falsification provides an additional boundary. Shuffled-pair controls could suppress scanner recovery as much as, or more than, true pairs, but they degraded same-region agreement and retrieval. Scanner suppression alone is therefore insufficient evidence; correct matching matters because it constrains what the model is asked to preserve.

5.2 Corrected Canine Fixed-Estimand Audit

Table 2 gives five-fold means from the no-training v2 adjudication. Original features retain descriptive category structure but make scanner identity highly recoverable. Centroid/QR and paired-linear corrections sharply reduce scanner recovery while maintaining category balanced accuracy close to the original and retaining high retrieval. Neural B32 attains the lowest mean scanner balanced accuracy in the table, but its retrieval is substantially lower. B64 recovers some retrieval while also increasing scanner recoverability. Under the registered three-axis rule, neither neural representation establishes a feature-space increment over every simple baseline. The B64-minus-B32 contrast makes the allocation trade-off explicit. Corrected category balanced accuracy changed by -0.0076 with interval $[-0.0194, +0.0071]$. Scanner balanced accuracy increased by $+0.0489$ ($[+0.0419, +0.0557]$), overall retrieval increased by $+0.0515$ ($[+0.0447, +0.0602]$), and worst-pair retrieval increased by $+0.0497$ ($[+0.0442, +0.0551]$). A wider tissue bottleneck therefore carried more region identity and more scanner information, without supported category gain.

The explicit acquisition branch retained strong scanner information in the corrected audit. This supports information routing rather than unaccounted erasure, but

Table 1: Fold-aware PA-NF minus equal-parameter two-branch control on SCORPION. Negative scanner-probe differences and positive retention differences are favorable. Retrieval is judged against a 0.02 noninferiority margin.

Metric	Mean difference	95% interval	Folds favorable	Unit count	Interpretation
Tissue scanner balanced accuracy	-0.3108	[-0.3346, -0.2858]	5/5	48 slides	Lower recoverability
Mean paired cosine	+0.03639	[+0.03355, +0.03954]	5/5	48 slides	Positive geometry contrast
Worst paired cosine	+0.03557	[+0.03167, +0.04064]	5/5	48 slides	Positive geometry contrast
Mean top-1 retrieval	+0.00004	[-0.00009, +0.00026]	1/5	48 slides	Noninferior, not improved
Worst-pair top-1 retrieval	0.00000	[-0.00085, +0.00085]	1/5	48 slides	Noninferior, not improved
Acquisition scanner balanced accuracy	+0.1180	[+0.1003, +0.1358]	5/5	48 slides	Stronger explicit routing

Table 2: Corrected five-fold canine SCC representation audit. Values are fold means. Category balanced accuracy is descriptive and not a clinical endpoint.

Representation	Category BA $\uparrow$	Scanner BA $\downarrow$	Overall retrieval $\uparrow$	Worst-pair retrieval $\uparrow$	Role
Original frozen features	0.4439	0.8635	0.8724	0.7613	Uncorrected reference
Centroid/QR scanner projection	0.4414	0.3278	0.9099	0.8196	Strong simple correction
Paired linear scanner transform	0.4422	0.3687	0.9022	0.8283	Paired simple correction
PCA component removal	0.3949	0.8157	0.9197	0.8512	Weak scanner suppression
PA-NF tissue branch B32	0.4331	0.2996	0.7192	0.5848	Neural reference
PA-NF tissue branch B64	0.4255	0.3486	0.7707	0.6345	Parameter-matched wider branch

does not establish that the acquisition branch is biologically pure or that factors can be swapped to produce valid images.

5.3 Cross-Backbone and Synthetic Boundaries

The frozen SCORPION objective was also evaluated on DINOv2-Base, Phikon, and ImageNet ResNet50 feature substrates without backbone-specific objective tuning. Across those representation families, the tissue branch reduced linear scanner recoverability while preserving near-saturated same-region retrieval. These transfers are supportive robustness checks; the primary inferential comparison remains the capacity-matched DINOv2 campaign.

Controlled synthetic studies with known latent factors examine paired exposure, anchor diversity, and bottleneck capacity. They are useful for identifying mechanisms under specified generators, but their accessibility effects did not transport to corrected real canine category accessibility. Synthetic recovery is therefore not treated as biological validation.

6 Interpretation

The combined evidence supports a narrower conclusion than the original project framing. Paired acquisition is useful because it supplies a constraint on what should remain shared across scanner views. PA-NF can use that constraint to lower scanner recoverability while preserving same-region retrieval on SCORPION and to retain scanner information in an explicit acquisition branch. However, the corrected canine audit shows that a neural factorization is not automatically better than linear scanner removal. Strong simple baselines can occupy a better scanner/category/retrieval trade-off.

This distinction matters. “Scanner invariant” is not equivalent to “biological,” and high same-region retrieval is not equivalent to preserved disease information. The representations should be interpreted operationally: one branch is trained to be more tissue-oriented, the other

more acquisition-oriented, and the observed allocation is partial and protocol dependent.

7 Limitations

The study has several important limitations.

1. SCORPION contains only 48 original human slides, even though it provides 2,400 aligned patches. The slide, not the patch, is the inferential unit.
2. The canine dataset is an independent multi-scanner geometry but differs in species, tissue distribution, resolution, and label construction. It is not a human clinical validation cohort.
3. Linear probes measure recoverability under a specified estimator; they do not prove information-theoretic independence.
4. Cosine agreement and retrieval are representation-geometry metrics. They do not prove biological preservation or diagnostic equivalence.
5. The main experiments operate on frozen feature vectors. They do not establish pixel-space normalization, valid scanner swapping, or end-to-end deployment behavior.
6. The objective contains several interacting terms. Ablations localize important effects but do not establish universal necessity or causal uniqueness.
7. Broader biological claims require additional independently collected human paired-scanner cohorts and clinically meaningful endpoints.

8 Evidence and Reproducibility

The corrected evidence is maintained as forward-valid, immutable release packages. The central paired-acquisition release is bound by SHA-256 and records the claim boundary, dataset manifests, commands, source commits, and output hashes. The capacity-matched SCORPION package records 175 unique completed runs, seven variants, five folds, five seeds, and 36 aggregate contrast rows. The canine v2

adjudication is bound to a 438,446-byte result artifact whose SHA-256 begins 2cccaad7aab89586; the complete digest is recorded in the release manifest.

Code, tests, evidence manifests, and the manuscript source are available at the repository linked on the first page. The authoritative public claim boundary overrides older PDFs, captions, and reports whenever they conflict. The earlier focused PDF remains superseded; this corrected manuscript does not retroactively validate it.

9 Conclusion

PA-NF demonstrates that matched acquisitions can supervise an explicit allocation of scanner-associated and tissue-associated information in frozen pathology embeddings. Capacity-matched SCORPION evidence supports lower linear scanner recoverability with noninferior same-region retrieval and strong acquisition-branch scanner retention. The corrected canine study adds the essential negative result: the neural feature space did not establish an increment over strong simple corrections, and a wider tissue bottleneck traded higher retrieval for higher scanner recoverability without category gain. The scientifically defensible conclusion is partial structured separation under the tested conditions, not pure disentanglement, scanner-free biology, or clinical utility.

Data and Code Availability

SCORPION is available from Zenodo under DOI [10.5281/zenodo.16517924](https://doi.org/10.5281/zenodo.16517924). The multi-scanner canine SCC dataset is available from Zenodo under record [7418555](https://doi.org/10.5281/zenodo.7418555). Reproduction code, corrected evidence manifests, tests, and publication source are available at [the project repository](#).

Ethics and Intended Use

This work analyzes public, de-identified research datasets and frozen representations. It is research only. It is not clinical software, a medical device, diagnostic validation, or deployment guidance.

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